

CONTACT Computational Imaging
INFORMATION Biohub SF

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RESEARCH AND
WORK
EXPERIENCE

Computational Imaging - Biohub

Aug 2024 – present

R&D Engineer - Developing high-throughput image analysis and smart microscopy solutions for dynamic imaging. Recent work on registration, tracking, and segmentation of infected cells reported in [1–3] tracking of live dynamics samples during microscopy imaging acquisition [4], and evaluation of dynamic 3D virtual staining of live cells [5].

LIDS - Laboratory of Image Data Science

Aug 2021 – 2025

Graduate Researcher - Developing an automatic detector of image descriptors in 4D (3D+t) DCE-MRI of the thorax for the diagnosis of malignant pleural mesothelioma cancer. Using pytorch Unet and diffusion models for data augmentation, we perform automatic segmentation of the region containing the tumor. Within it, we detect possible tumors using a graph-based supervoxel algorithm [6–9].

PREDICT-Meso Network

Feb 2023 – Jul 2023

Visiting Researcher - Developed an interactive segmentation tool for 4D DCE-MR images of patients suspected of Malignant Pleural Mesothelioma for label ground truth purposes [10].

LIII - Lab of Image, Intelligence and Innovation

Jan 2020 – Apr 2020

Visiting Student Researcher - Learned to use CNN in TensorFlow for MRI spine segmentation.

LAB2I - Automation and Intelligent Instrumentation Laboratory

Aug 2017-Dec 2019

Undergraduate Research Internship - Developed an embedded system with sensors to monitor the parameters of temperature and electric current of water treatment by electrocoagulation process [11, 12].

Peapply & Co. Data Engineer

May 2021 – Oct 2021

Designing and building systems for collecting, storing, and analyzing data at scale in Pyspark and SQL.

EDUCATION

University of Campinas SP, BR

Aug 2021 – 2025

M.Sc. in Computer Science

- Thesis: Towards Automatic Detection of Pleural Mesothelioma Biomarker in 4D Dynamic MR Imaging [7]
- GPA: 4.0/4.0

State University of Londrina PR, BR

Apr 2016 - Mar 2021

B.Sc. in Electrical Engineering

- Thesis: Study of Deep Learning and its Application to Medical Images

ADVANCED
TRAINING

Marine Biological Laboratory (MBL), Woods Hole, MA, USA

2025

Analytical and Quantitative Light Microscopy

Advanced intensive research training course focused on quantitative fluorescence microscopy, optical physics, image analysis, and computational methods for biological imaging.

COURSEWORK

Graduate Digital Image Processing, Image Analysis, Parallelization and Optimization Algorithms for Machine Learning Graphs, Machine Learning and Pattern Recognition, Deep Learning, Graph Algorithms.

Undergraduate Digital Circuits, Analog and Digital Electronics, C programming, Embedded Systems, Digital Signal Processing, Modern Programming Languages, Communication Principles, Electronic Instrumentation, Microelectronics (ARM), Biomedical Engineering, VHDL.

TEACHING
ASSISTANT

MC202 - Data Structure

Aug 2022 - Nov 2022

MC832 - Computer Network

Mar 2022 - Jun 2022

VOLUNTEER
EXPERIENCE

Techninas - Girls in Tech

2018 - 2020

Student Branch IEEE UEL

2016 - 2018

Women in Engineering IEEE UEL

2016 - 2018

ISBI 2026 - IEEE International Symposium on Biomedical Imaging Meso-Regions: 4D Segmentation of Early Contrast Enhancement Biomarkers for Pleural Mesothelioma Diagnosis [6].
SPIE BiOS 2026 DynaTrack: label-agnostic smart tracking for mapping dynamics of organ development [4].
iMig 2023 - 16th International Conference of the International Mesothelioma interest group Development of an automated machine learning tool for reporting of MRI-Early Contrast Enhancement in Mesothelioma [2].
SIBGRAPI 2023 - 36th Conference on Graphics, Patterns and Images Workshop - Superpixel segmentation: from theory to applications [8].

- [1] S. Pradeep, A. Imran, Z. Liu, T. M. Theodoro, E. Hirata-Miyasaki, I. Ivanov, M. Bhavé, S. Khadka, H. Woosley, C. Arias, and S. B. Mehta, “Contrastive learning of cell state dynamics in response to perturbations,” 2024.
- [2] E. Hirata-Miyasaki, S. Pradeep, Z. Liu, A. Imran, T. M. Theodoro, I. Ivanov, S. Khadka, M. Bhavé, H. Woosley, C. Arias, *et al.*, “Mapping cell state dynamics with ai-enabled live cell imaging,” in *High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond X*, p. PC133300R, SPIE, 2025.
- [3] E. Hirata-Miyasaki, S. Pradeep, Z. Liu, A. Imran, T. M. Theodoro, I. E. Ivanov, S. Khadka, S.-C. Lee, M. E. Grunberg, H. Woosley, M. Bhavé, C. Arias, and S. B. Mehta, “Embedding cell state dynamics via contrastive learning of representations of 3d dynamic imaging datasets,” in *NeurIPS 2025 Workshop for Imageomics: Discovering Biological Knowledge from Images Using AI*, 2025.
- [4] T. M. Theodoro, A. Q. Ryan, R. Banks, E. Hirata-Miyasaki, T. Lao, S. Jaisingh, A. Jacobo, S. B. Mehta, and I. Ivanov, “Dynatrack: label-agnostic smart tracking for mapping dynamics of organ development,” in *High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond XI*, vol. 13862, pp. 23–31, SPIE, 2026.
- [5] A. A. Kalinin, D. Zheng, T. M. Theodoro, I. E. Ivanov, E. Hirata-Miyasaki, S.-C. Lee, Z. Liu, S. Varra, T. Chandler, S. Pradeep, A. Liu, M. Leonetti, A. Arias, B. Huang, and S. Mehta, “Dynacell: An evaluation framework for dynamic 3D virtual staining of live cells,” in *The Fortieth Annual Conference on Neural Information Processing Systems Evaluations and Datasets Track*, 2026. Poster; accepted.
- [6] T. M. Theodoro, I. F. Silva, S. Tsim, K. Blyth, and A. X. Falcao, “Meso-regions: 4d segmentation of early contrast enhancement biomarkers for pleural mesothelioma diagnosis,” in *2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, pp. 1–5, 2026.
- [7] T. M. Theodoro, “Towards automatic detection of pleural mesothelioma biomarker in 4d dynamic mr imaging,” master’s thesis, University of Campinas (UNICAMP), Institute of Computing, Campinas, Brazil, 2025.
- [8] I. Borlido, F. Belém, L. Melo, T. M. Theodoro, I. F. Silva, Z. K. G. do Patrocínio Jr, A. X. Falcao, and S. J. F. Guimaraes, “Superpixel segmentation: from theory to applications,” in *(to appear in) 2023 36th SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)*, 2023.
- [9] S. G. Armato, S. I. Katz, T. Frauenfelder, G. Jayasekera, A. Catino, K. G. Blyth, T. Theodoro, P. Rousset, K. Nackaerts, and I. Opitz, “Imaging in pleural mesothelioma: A review of the 16th international conference of the international mesothelioma interest group,” *Lung Cancer*, vol. 193, p. 107832, 2024.
- [10] I. F. da Silva and T. M. Theodoro, “LIDS-UNICAMP: MISe: Multidimensional Image Segmentation Software (initial alpha release),” Aug 2023.
- [11] T. Theodoro, “Study an Development of an Automation System to Water Treatment by Electrocoagulation Process,” in *Congress proceedings.*, (Londrina), Annual Congress of Scientific Initiation of UEL, ISSN 2447-4118, 2018.
- [12] T. Theodoro, “Improvement of an Automation System To Water Treatment by Eletrocoagulation Process,” in *Congress proceedings.*, (Londrina), Annual Congress of Scientific Initiation of UEL, ISSN 2447-4118, 2019.